Reg. No.: …… Name: ……..

ALBERTIAN INSTITUTE OF SCIENCE & TECHNOLOGY

Diploma Second Internal Examination (September 2025)

Program Code and Semester: AM S3

Course Code and Name: 3341 DISCRETE MATHEMATICS

Max. Marks: 20		**Duration: 1 Hours**		
Q. No	**Question**	**Marks**	**CO**	**BL**
Part A **Answer all questions, each question carries 1 mark**				
1.	State Pigeon hole Principle.	**1**	**3**	**R**
2.	Define injective function.	**1**	**2**	**R**
Part B **Answer all questions, each question carries 3 marks**				
3.	If $A = \{1, 2, 3, 4\}$, give an example of a relation on A which is reflexive and transitive, but not symmetric.	**3**	**2**	**A**
4.	Let set $X = \{1, 2, 3, 4\}$ and relation $R = \{(x, y) \mid x > y\}$. Draw the graph of R and also give its matrix.	**3**	**2**	**U**
Part C **Answer any 2 questions, each question carries 6 marks**				
5.	An urn contains 15 balls, 8 of which are red and 7 are black. In how many ways can 5 balls be chosen so that (i) all the five are red (ii) 2 are red and 3 are black (iii) At least 3 are red.	**6**	**3**	**A**
6.	Explain different types of relations in sets with examples.	**6**	**2**	**U**
7.	a. Find the number of permutations of the letters of the word INDEPENDENCE. b. In how many ways can the letters of the word 'MATHEMATICS' be arranged such that vowels must always come together?	**6**	**3**	**A**